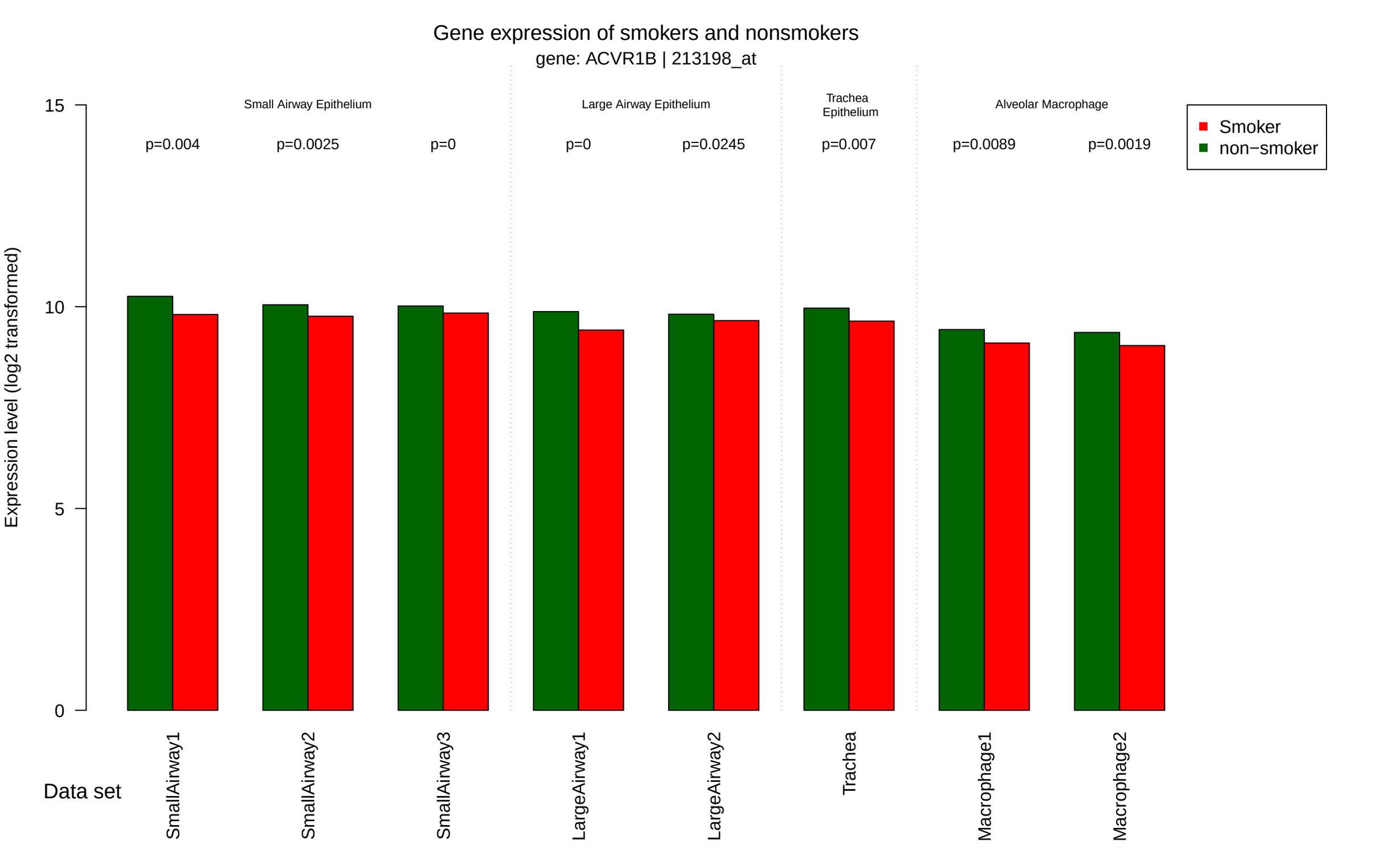
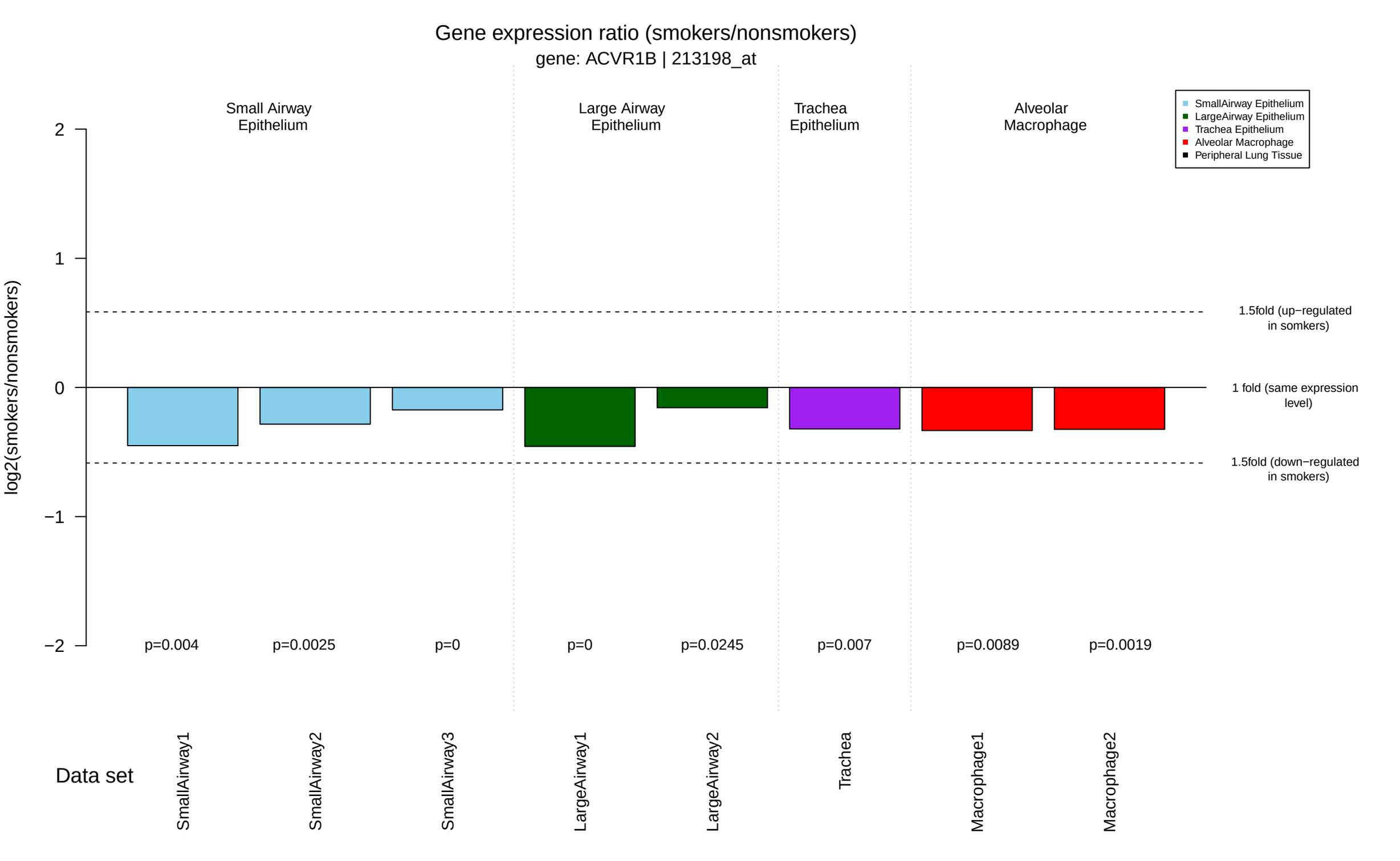
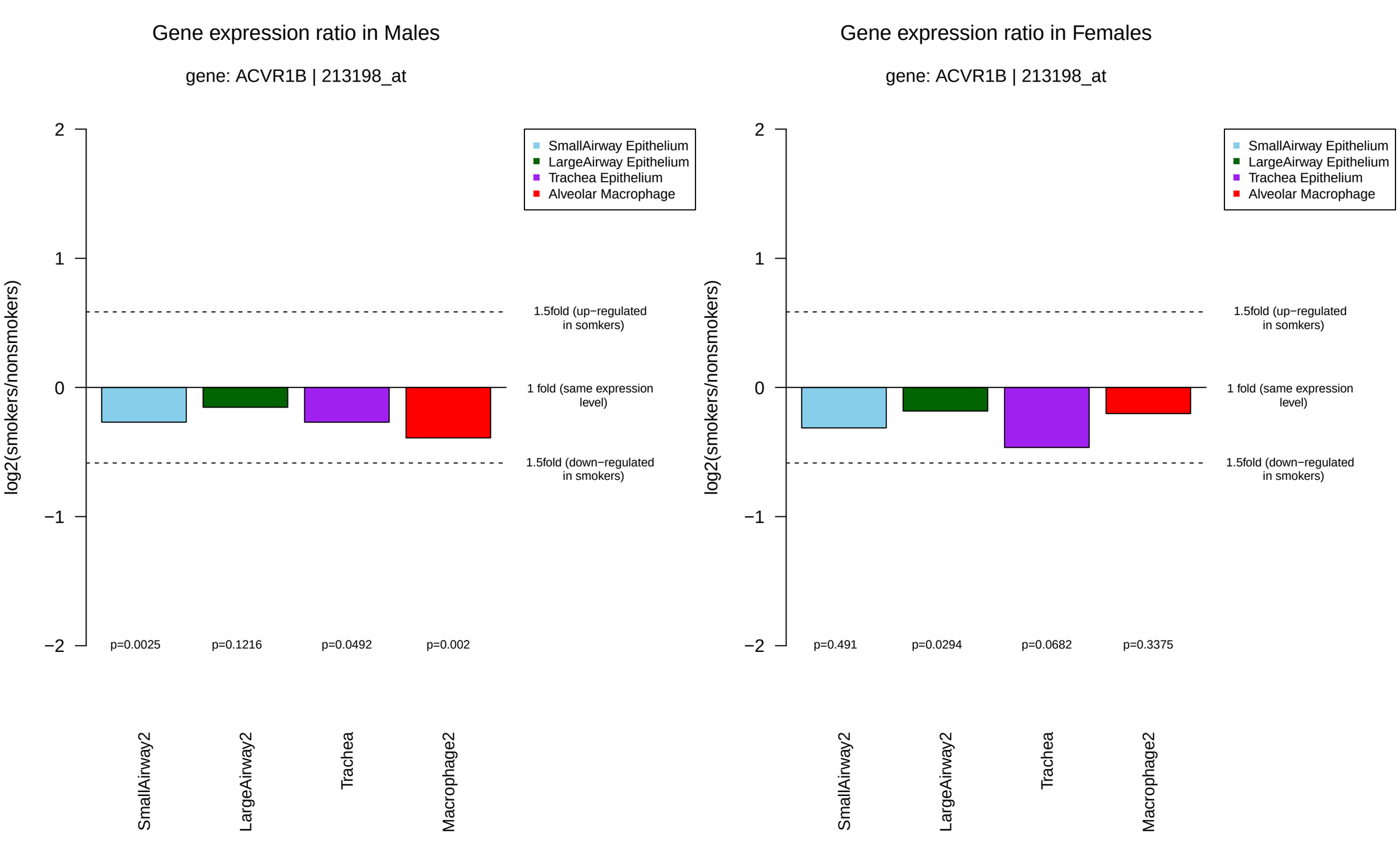
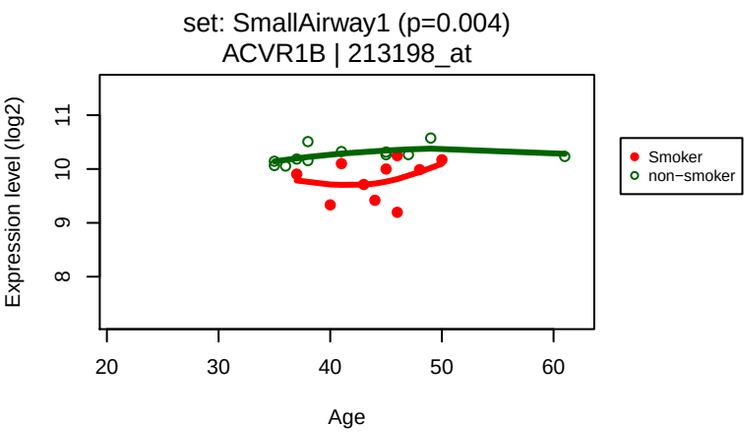
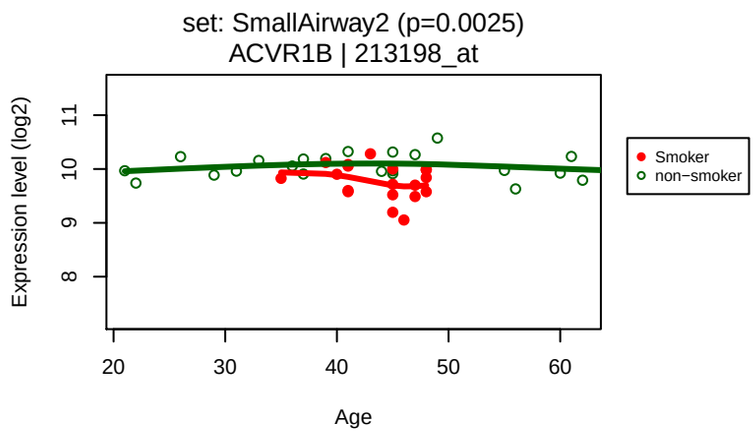
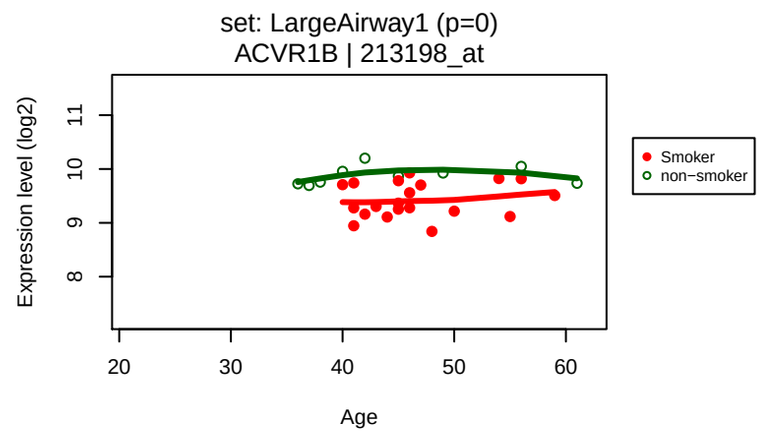

Gene expression of smokers and nonsmokers
gene: ACVR1B | 213198_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.004
p=0.0025
p=0
p=0
p=0.0245
p=0.007
p=0.0089
p=0.0019
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

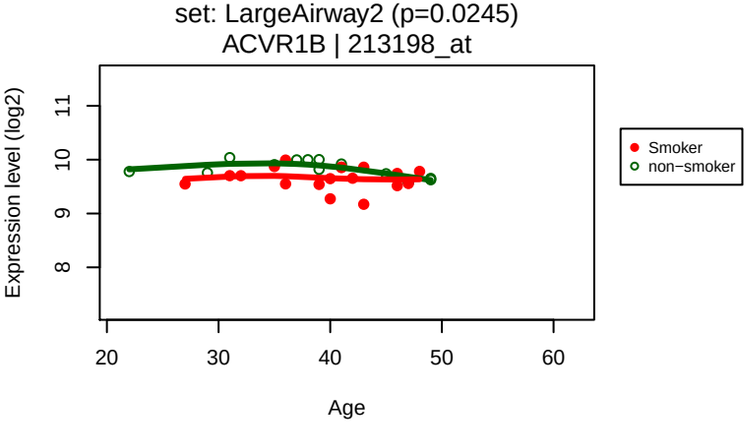

Gene expression ratio (smokers/nonsmokers)
gene: ACVR1B | 213198_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.004
p=0.0025
p=0
p=0
p=0.0245
p=0.007
p=0.0089
p=0.0019
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

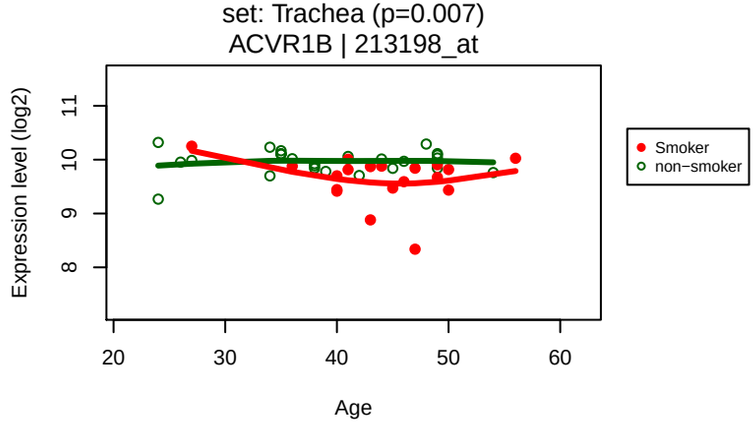

Gene expression ratio in Males
gene: ACVR1B | 213198_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0025
p=0.1216
p=0.0492
p=0.002
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACVR1B | 213198_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.491
p=0.0294
p=0.0682
p=0.3375
SmallAirway2
LargeAirway2
Trachea
Macrophage2

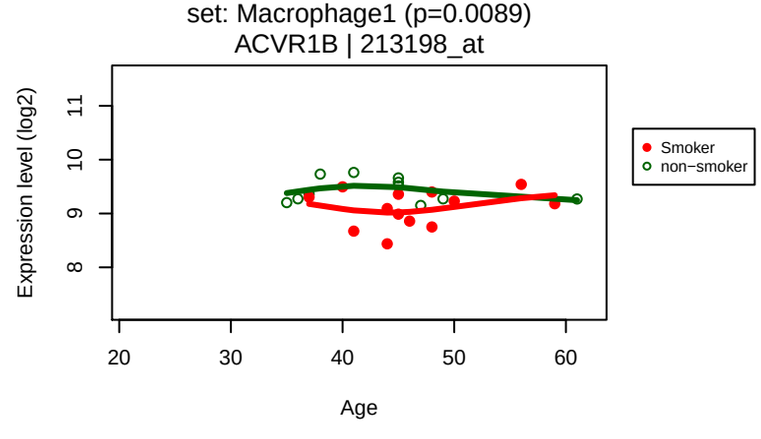

set: SmallAirway1 (p=0.004)
ACVR1B | 213198_at
Expression level (log2)
Age
Smoker
non−smoker

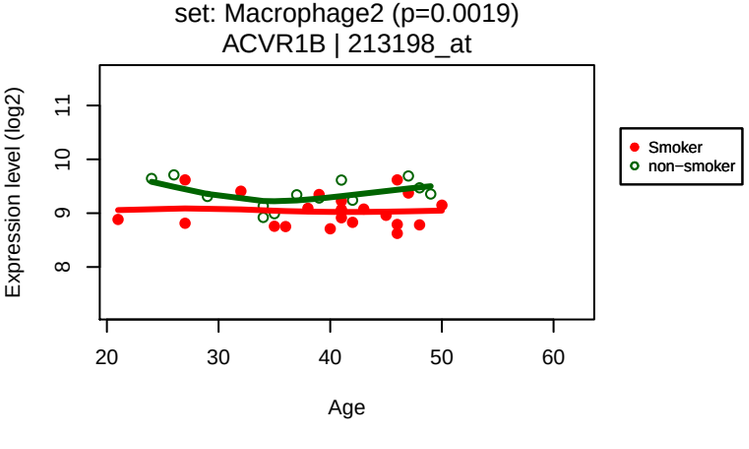

set: SmallAirway2 (p=0.0025)
ACVR1B | 213198_at
Expression level (log2)
Age
Smoker
non−smoker

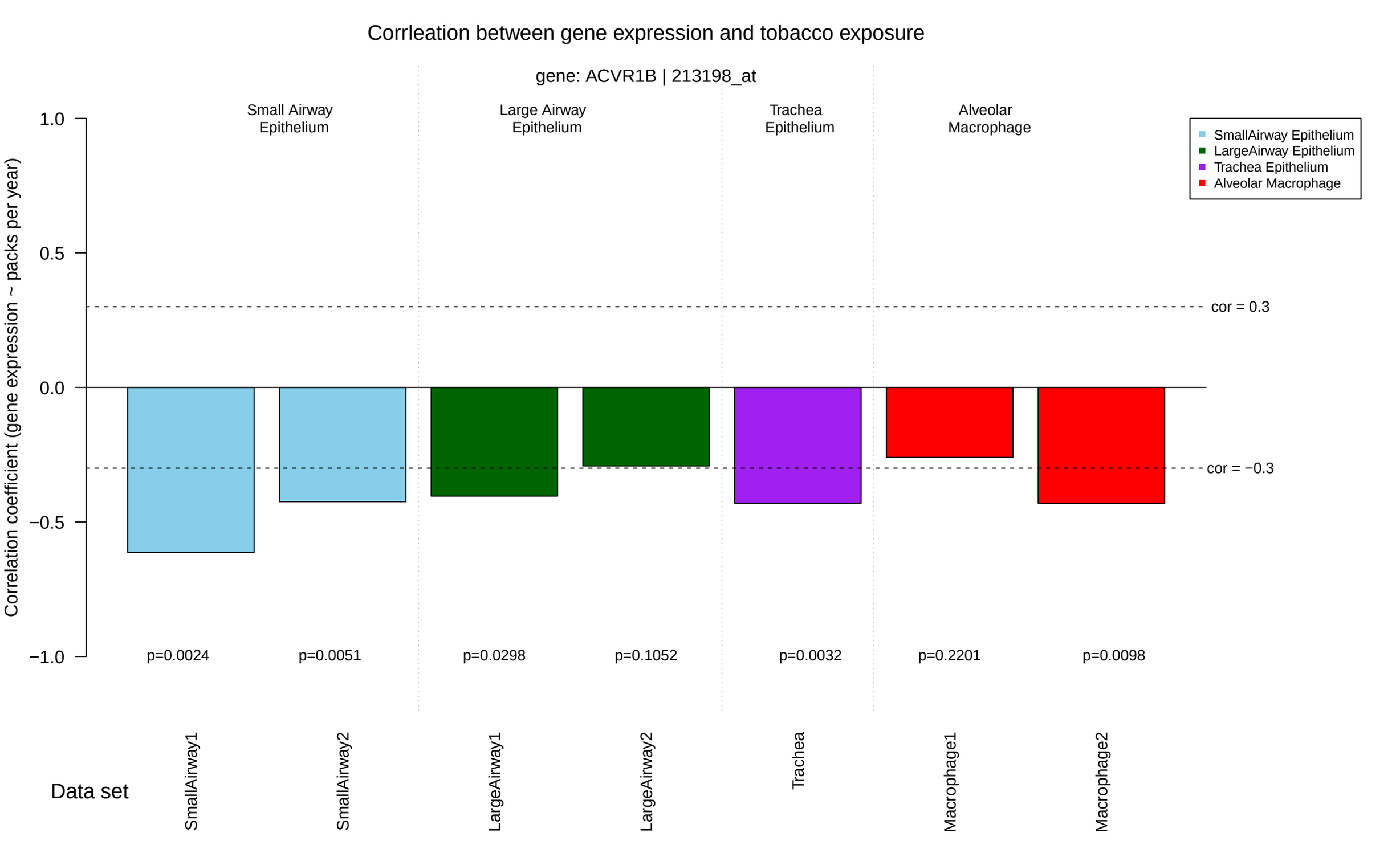

set: LargeAirway1 (p=0)
ACVR1B | 213198_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0245)
ACVR1B | 213198_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.007)
ACVR1B | 213198_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0089)
ACVR1B | 213198_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0019)
ACVR1B | 213198_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACVR1B | 213198_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0024
p=0.0051
p=0.0298
p=0.1052
p=0.0032
p=0.2201
p=0.0098
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2